

## Scope

The SAMA7G5 Series device that you have received conforms functionally to the current SAMA7G5 Series device data sheet (DS60001765) or SAMA7G5 Series System-in-Package (SiP) data sheet (DS50003577), except for the anomalies described in this document.

The silicon issues discussed in the following pages are for silicon revisions with the device revision and device identification listed in the following table. The silicon issues are summarized in the [Silicon Issue Summary](#).

Data sheet clarifications and corrections (if applicable) are located in the [Data Sheet Clarifications](#) section.

The silicon device revisions and device IDs are shown in the following table.

**Table 1.** SAMA7G5 Series Device Identification

| Ordering Code             | Device Revision | Device Identification |
|---------------------------|-----------------|-----------------------|
|                           |                 | CHIPID_CIDR[31:0]     |
| SAMA7G54(T)-V/4HB         | A0              | 0x80162110            |
| SAMA7G54(T)-V/4HB         | A1              | 0x80162111            |
| SAMA7G54(T)-V/7EW         |                 |                       |
| SAMA7G54(T)-E/4HB(VAO)    |                 |                       |
| SAMA7G54D1G(T)-I/4UB      | A1-D1G          |                       |
| SAMA7G54D1G(T)-V/XHB      |                 |                       |
| SAMA7G54D1G(T)-E/XHB(VAO) |                 |                       |
| SAMA7G54D2G(T)-I/4UB      | A1-D2G          |                       |
| SAMA7G54D2G(T)-V/XHB      |                 |                       |
| SAMA7G54D2G(T)-E/XHB(VAO) |                 |                       |
| SAMA7G54D4G(T)-I/4UB      | A1-D4G          |                       |
| SAMA7G54D4G(T)-V/XHB      |                 |                       |
| SAMA7G54D4G(T)-E/XHB(VAO) |                 |                       |

**Note:** Refer to the “Chip Identifier (CHIPID)” and “Product Identification System” sections in the current device data sheet for detailed information on chip identification for your specific device.

## 1. Silicon Issue Summary

In this table and in subsequent sections, the following applies:

- “X” means the device revision is affected by the erratum.
- “–” means the device revision is not affected by the erratum.

**Table 1-1.** Silicon Issue Summary

| Module    | Erratum                                                                   | Affected Device Revisions |    |        |        |        |
|-----------|---------------------------------------------------------------------------|---------------------------|----|--------|--------|--------|
|           |                                                                           | A0                        | A1 | A1-D1G | A1-D2G | A1-D4G |
| ROM Code  | Boot failure on eMMC memories                                             | X                         | X  | X      | X      | X      |
| Cortex-A7 | PMU interrupt spurious rise                                               | X                         | X  | X      | X      | X      |
|           | AXIERRIRQ interrupt spurious rise                                         | X                         | X  | X      | X      | X      |
| EIC       | WPVS bit incorrect behavior                                               | X                         | X  | X      | X      | X      |
| XDMAC     | Data corrupted when number of AXI outstanding transactions differs from 1 | X                         | X  | X      | X      | X      |
|           | Some XDMAC0 and XDMAC1 channels ineffective                               | X                         | –  | –      | –      | –      |
| RSTC      | RSTC_SR.RSTTYP not showing GENERAL_RST                                    | X                         | X  | X      | X      | X      |
| RTC       | RTC_TSTR0 timestamping error                                              | X                         | X  | X      | X      | X      |
| CHIPID    | CHIPID_EXID may report a wrong value                                      | –                         | –  | X      | X      | –      |
| OTPC      | OTPC restricted operating range in Write mode                             | X                         | X  | X      | X      | X      |
|           | OTPC wrong default configuration                                          | X                         | X  | X      | X      | X      |
| PMC       | MCKRDY flag error                                                         | X                         | X  | X      | X      | X      |
|           | Delay to first establish PCK                                              | X                         | X  | X      | X      | X      |
|           | PCK and GCLK Ready status issue                                           | X                         | X  | X      | X      | X      |
|           | Processor (CPU_CLK0) and main system bus clock (MCK0) source selection    | X                         | X  | X      | X      | X      |
| PIO       | Open drain management limitation                                          | X                         | X  | X      | X      | X      |
| ADC       | Spurious effect when zeros written to ADC_EOC_IDR                         | X                         | X  | X      | X      | X      |
|           | EOC interrupts not disabled when ones written to ADC_EOC_IDR              | X                         | X  | X      | X      | X      |
|           | ADC_IMR interrupts enabled when ones written to ADC_EOC_IDR               | X                         | X  | X      | X      | X      |
|           | Temperature sensor still enabled when stopped without conversion          | X                         | X  | X      | X      | X      |
|           | Temperature sensor spurious activation with CH30                          | X                         | X  | X      | X      | X      |
|           | Sleep mode ineffective                                                    | X                         | X  | X      | X      | X      |
| CSI2DC    | Decompression modes 12-7-12, 12-8-12, 10-7-10 and 10-6-10 not compliant   | X                         | X  | X      | X      | X      |
| ISC       | Spurious DMA descriptor writing                                           | X                         | X  | X      | X      | X      |
|           | Incoming pixels corrupted after overload                                  | X                         | X  | X      | X      | X      |
|           | Frequency limitation                                                      | X                         | –  | –      | –      | –      |
| SSC       | Inverted left/right channels                                              | X                         | X  | X      | X      | X      |
|           | TD output delay                                                           | X                         | X  | X      | X      | X      |
| SPDIFRX   | SPDIFRX left/right inversion                                              | X                         | X  | X      | X      | X      |
| AES       | SPLIP mode does not work with some header sizes                           | X                         | X  | X      | X      | X      |

Table 1-1. Silicon Issue Summary (continued)

| Module  | Erratum                                                                                                             | Affected Device Revisions |    |        |        |        |
|---------|---------------------------------------------------------------------------------------------------------------------|---------------------------|----|--------|--------|--------|
|         |                                                                                                                     | A0                        | A1 | A1-D1G | A1-D2G | A1-D4G |
| SECUMOD | Dynamic detection intrusion (PIOBU) alarm issue                                                                     | X                         | X  | X      | X      | X      |
|         | Tamper timestamping polarity error                                                                                  | X                         | X  | X      | X      | X      |
|         | SECUMOD registers BMPR and WKPR reading issue                                                                       | X                         | X  | X      | X      | X      |
| GMAC    | GMAC0 not functional with multiple queues in 10/100 Half Duplex mode                                                | X                         | X  | X      | X      | X      |
|         | Incorrect reading of Specific Address filter registers on GMAC0 and GMAC1                                           | X                         | -  | -      | -      | -      |
|         | Incorrect reading of Type 1 Screener registers on GMAC0 and GMAC1                                                   | X                         | -  | -      | -      | -      |
|         | Incorrect reading of Type 2 Screener registers on GMAC0 and GMAC1                                                   | X                         | -  | -      | -      | -      |
|         | GTSUCOMP ineffective connection to TC1                                                                              | X                         | X  | X      | X      | X      |
| FLEXCOM | Write Protection ineffective on FLEXCOM8 to FLEXCOM11                                                               | X                         | X  | X      | X      | X      |
| SDMMC   | SDMMC failure when changing speed mode or performing ALL soft reset on the fly                                      | X                         | X  | X      | X      | X      |
|         | SDR104, HS200, HS400 modes are not functional                                                                       | X                         | X  | X      | X      | X      |
|         | GCLK cannot be stopped                                                                                              | X                         | X  | X      | X      | X      |
|         | SDHC blocked after switch from high-speed mode                                                                      | X                         | X  | X      | X      | X      |
| MCAN    | Edge filtering causes mis-synchronization when falling edge at Rx input pin coincides with end of integration phase | X                         | X  | X      | X      | X      |
|         | Configuration of MCAN_NBTP.NTSEG2 = '0' not allowed                                                                 | X                         | X  | X      | X      | X      |
|         | Retransmission in DAR mode due to lost arbitration at the first two identifier bits                                 | X                         | X  | X      | X      | X      |
|         | Tx FIFO message sequence inversion                                                                                  | X                         | X  | X      | X      | X      |
|         | Unexpected High Priority Message (HPM) interrupt                                                                    | X                         | X  | X      | X      | X      |
|         | Issue message transmitted with wrong arbitration and control fields                                                 | X                         | X  | X      | X      | X      |
|         | Debug message handling state machine not reset to Idle when CCCR.INIT is set                                        | X                         | X  | X      | X      | X      |
|         | Message order inversion when transmitting from dedicated Tx buffers configured with same message ID                 | X                         | X  | X      | X      | X      |
|         | Frame transmitted despite confirmed transmit cancellation for CAN-FD messages with more than 8 data bytes           | X                         | X  | X      | X      | X      |
|         | MCAN_TSU_TSCFG reset after read                                                                                     | X                         | X  | X      | X      | X      |
|         | MCAN_TSU_TSS1 not reset after a MCAN_TSU_TSx read                                                                   | X                         | X  | X      | X      | X      |
|         | MCAN_TSU_ATB read resets the timebase value                                                                         | X                         | X  | X      | X      | X      |
| TC      | TC0 Channel 2 registers incorrect reading                                                                           | X                         | -  | -      | -      | -      |
| UDPHS   | EHCI spurious stop when Suspend mode occurs on port A                                                               | X                         | X  | X      | X      | X      |

**Table 1-1. Silicon Issue Summary (continued)**

| Module          | Erratum                                 | Affected Device Revisions |    |        |        |        |
|-----------------|-----------------------------------------|---------------------------|----|--------|--------|--------|
|                 |                                         | A0                        | A1 | A1-D1G | A1-D2G | A1-D4G |
| Low Power Modes | <a href="#">ULP2 mode does not work</a> | X                         | X  | X      | X      | X      |

## 2. ROM Code

### 2.1. Boot failure on e.MMC memories

The device fails to load a bootstrap program (boot.bin) from an e.MMC **USER** partition.

#### Work Around

Always use the e.MMC **BOOT** partition to store the boot.bin file and enable the e.MMC **BOOT** partition feature.

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 3. Cortex-A7 Processor (Arm)

#### 3.1. PMU interrupt spurious rise

The Performance Monitoring Unit (PMU) interrupt rises as soon as enabled.

##### Work Around

Do not enable the PMU interrupt.

##### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

#### 3.2. AXIERRIRQ interrupt spurious rise

The AXIERRIRQ interrupt rises as soon as enabled.

##### Work Around

Do not enable the AXIERRIRQ interrupt.

##### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 4. External Interrupt Controller (EIC)

### 4.1. WPVS bit incorrect behavior

The EIC\_WPSR.WPVS bit never rises to 1.

#### Work Around

None

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 5. DMA Controller (XDMAC)

### 5.1. Data corrupted when number of AXI outstanding transactions differs from 1

Data corruption may occur when the number of AXI outstanding transactions differs from 1.

#### Work Around

Limit to 1 the number of AXI outstanding transactions to access the AHB part in NICGPV. Performance of multichannel DMA transfers to the AHB part (SRAM, EBI, QSPI) is slightly impacted. Performance of transfers to the DDR memory is not affected. Apply the following settings:

```
NICGPV->NICGPV_AMIB[6].NICGPV_AMIB_FN_MOD = 0x3;
NICGPV->NICGPV_AMIB[13].NICGPV_AMIB_FN_MOD = 0x3;
```

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 5.2. Some XDMAC0 and XDMAC1 channels ineffective

Reading any register of channels 0, 1, 18, 19, 26 and 27 of XDMAC0 and XDMAC1 may return corrupted values.

#### Work Around

Do not use XDMAC0 and XDMAC1 channels 0, 1, 18, 19, 26 and 27.

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | -  | -      | -      | -      |  |  |  |

## 6. Reset Controller (RSTC)

### 6.1. RSTC\_SR.RSTTYP not showing GENERAL\_RST

In the Status register (RSTC\_SR), the RSTTYP field shows BACKUP\_RST instead of GENERAL\_RST.

#### Work Around

None

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 7. Real-time Clock (RTC)

### 7.1. RTC\_TSTR0 timestamping error

RTC\_TSTR0.TEVCNT fails to report the correct number of tamper event occurrences.

#### Work Around

None

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 8. Chip Identifier (CHIPID)

### 8.1. CHIPID\_EXID may report a wrong value

The CHIPID\_EXID register for SAMA7G54D1G and SAMA7G54D2G may report a wrong value for devices with lot traceability code 2306W9M and 2330WHM, respectively (refer to the section “Marking” of the data sheet).

The value reported is 0 instead of 0x00000018 (SAMA7G54D1G) and 0x00000020 (SAMA7G54D2G).

#### Work Around

None

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| -  | -  | X      | X      | -      |  |  |  |

## 9. OTP Controller (OTPC)

### 9.1. OTPC restricted operating range in Write mode

The write operations in the OTPC cannot be performed over the full temperature and VDDIN33 power supply ranges specified.

#### Work Around

The write operations in the OTPC are restricted to the following ambient temperature and VDDIN33 power supply ranges:

- $T_A = [0^{\circ}\text{C to } 50^{\circ}\text{C}]$
- $VDDIN33 = [3.15\text{V to } 3.6\text{V}]$

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 9.2. OTPC wrong default configuration

The default configuration of the OTPC cannot be used to access the OTP memory in Write mode.

#### Work Around

Prior to any write operation in the OTPC, the OTPC must be configured using the following code. This operation needs to be performed only once before the first write operation and whenever the peripheral reset (signal periph\_nreset) is asserted.

```
#define ARRAY_SIZE(a) (sizeof(a) / sizeof((a)[0]))

/*
 * writing one word lasts 350us
 * the timeout was chosen to be enough for writing 10 words  */
#define TIMEOUT 500000
#define OTPC_0 (0x1u << 0)
#define OTPC_1 16
#define OTPC_2 (0xffffu << OTPC_1)
#define OTPC_3 (0x4391u << OTPC_1)

static void otp_sama7g5_fixup(void)
{
    static const uint32_t fixup0[4] = {0x04194801, 0x01000000, 0x00000008, 0x00000000};
    static const uint32_t fixup1[4] = {0xf164801, 0x4c017d12, 0x02120e01, 0x00004000};
    __IO uint32_t *OTPC_4 = (__IO uint32_t *) ((uint8_t *)OTPC + 0x090);
    __IO uint32_t *OTPC_5 = (__IO uint32_t *) ((uint8_t *)OTPC + 0x0A0);
    __IO uint32_t *OTPC_6 = (__IO uint32_t *) ((uint8_t *)OTPC + 0x0B0);
    uint32_t timeout;
    int i;

    timeout = TIMEOUT;
    *OTPC_4 = OTPC_0 | OTPC_3;
    while (!(OTPC->OTPC_SR & OTPC_SR_UNLOCK) && --timeout > 0);

    for (i = 0; i < ARRAY_SIZE(fixup0); i++)
        OTPC_5[i] = fixup0[i];

    for (i = 0; i < ARRAY_SIZE(fixup1); i++)
        OTPC_6[i] = fixup1[i];

    timeout = TIMEOUT;
    *OTPC_4 = OTPC_3;
    while ((OTPC->OTPC_SR & OTPC_SR_UNLOCK) && --timeout > 0); }
```

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 10. Power Management Controller (PMC)

### 10.1. MCKRDY flag error

When PMC\_CPU\_CKR.MDIV is greater than 1, if the PMC\_CPU\_CKR.CSS field is modified, the MCKRDY signal may be stuck at 0.

#### Work Around

Use a software timeout of 64 cycles of CPU clock instead of polling the MCKRDY bit.

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 10.2. Delay to first establish PCK

When enabling a PCK after a reset, the delay before establishing the PCK with the correct frequency is 255 cycles of the PCK source clock. Once this delay has elapsed, and as long as the core reset is not asserted, there is no more additional delay when disabling/enabling the PCK.

#### Work Around

None

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 10.3. PCK and GCLK Ready status issue

The PCK and GCLK Ready signals are only affected by the enable/disable of the corresponding clock (PMC\_SCER.PCKx, PMC\_SCDR.PCKx or PMC\_SR.GCLKEN).

A Ready signal at '1' does not imply the clock is correctly established with the required frequency, hence the Ready status is not affected by the modification of the source or the dividing ratio of the clock. This means that:

1. modifying PMC\_PCKx.CSS or PMC\_PCKx.PRES does not make PMC\_SR.PCKRDYx fall,
2. modifying PMC\_PCR.GCLKCSS or PMC\_PCR.GCLKDIV does not make PMC\_SR.GCLKRDY fall.

#### Work Around

None

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 10.4. Processor (CPU\_CLK0) and main system bus clock (MCK0) source selection

When changing the fields CSS or CPCSS in the CPU Clock register (PMC\_CPU\_CKR) from any PLL source clocks (PLLxCKx) to Slow Clock source (SLOW\_CLK), the clock switching circuitry first switches from the PLL source to MAINCK source, then to Slow Clock source.

There is no impact on the clock switching sequence or device behavior. This intermediate step can be observed when the main system bus clock is output on a PCK pin.

#### Work Around

None

### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 11. Parallel Input/Output Controller (PIO)

### 11.1. Open drain management limitation

PIOC does not allow open drain configuration (PIO\_CFGRx.OPD=1) when a peripheral is selected (PIO\_CFGRx.FUNC different from 0).

TWI/TWIHS peripherals are not affected.

#### Work Around

None

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 12. Analog-to-Digital Converter (ADC) Controller

### 12.1. Spurious effect when zeros written to ADC\_EOC\_IDR

Writing 0s to ADC\_EOC\_IDR enables EOC interrupts instead of having no effect.

#### Work Around

None

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 12.2. EOC interrupts not disabled when ones written to ADC\_EOC\_IDR

Writing 1s to ADC\_EOC\_IDR does not disable the EOC interrupts as it should.

#### Work Around

The channels can be disabled from ADC\_CHDR if unused.

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 12.3. ADC\_IMR interrupts enabled when ones written to ADC\_EOC\_IDR

Writing 1s to ADC\_EOC\_IDR enables interrupts in ADC\_IMR. If interrupts are pending in ADC\_ISR, an interrupt is triggered to the interrupt controller.

Writing to ADC\_EOC\_IDR is not recommended.

#### Work Around

Immediately after writing 1s to ADC\_EOC\_IDR, write 1s to ADC\_IDR to disable the unwanted interrupts (store previous value, disable all, reenable previous value).

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 12.4. Temperature sensor still enabled when stopped without conversion

The temperature sensor remains active even when ADC\_TEMPMR.TEMPON is set to 0.

#### Work Around

To stop the temperature sensor and save its power consumption, perform a conversion prior to writing ADC\_TEMPMR.TEMPON=0.

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 12.5. Temperature sensor spurious activation with CH30

Enabling ADC channel 30 enables the temperature sensor.

**Work Around**

To stop the temperature sensor and save its power consumption, perform a conversion prior to writing ADC\_TEMP\_MR.TEMPON=0.

**Affected Device Revisions**

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

**12.6. Sleep mode ineffective**

Even if Sleep mode is selected by setting ADC\_MR.SLEEP, ADC does not enter Sleep mode after a conversion.

**Work Around**

Reset the ADC Controller with ADC\_CR.SWRST.

**Affected Device Revisions**

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 13. CSI-2 Demultiplexer Controller (CSI2DC)

### 13.1. Decompression modes 12-7-12, 12-8-12, 10-7-10 and 10-6-10 not compliant

When CSI2DC\_VPCFGR.DE (Decompression Enable) is set to 1, the following RAW Data Decompression modes are not supported: CSI2DC\_VPCFGR.DM = 0, 1, 4, 5.

#### Work Around

None

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 14. Image Sensor Controller (ISC)

### 14.1. Spurious DMA descriptor writing

Due to AXI transaction reordering, DMA descriptors writing may occur before the last image is written in the memory, even if the DONE flag is set. The user cannot read the DMA descriptors before the full image is written.

#### Work Around

1. Poll the bit ISC\_INTSR.DDONE.
2. Perform an extra read of ISC\_INTSR to enable DDR Controller writing.
3. Poll the bit UDDRC\_PSTAT.WR\_PORT\_BUSY\_3 with:  

```
while (DDRUMCTL_REGS->UDDRC_PSTAT & UDDRC_PSTAT_WR_PORT_BUSY_3(1));
```

Once done, the full image is written in the memory and the DMA descriptors can be written.

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 14.2. Incoming pixels corrupted after overload

In case of overload, incoming pixels may be corrupted.

#### Work Around

Discard the last frame when the ISC\_INTSR.DAOV bit is set.

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 14.3. Frequency limitation

Timing issues may occur in harsh conditions (low voltage and/or high temperature).

#### Work Around

In harsh conditions, limit the operating frequency to  $f_{MCK3} < 184$  MHz.

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | -  | -      | -      | -      |  |  |  |

## 15. Synchronous Serial Controller (SSC)

### 15.1. Inverted left/right channels

When the SSC is in Client mode, the TF signal is derived from the codec and not controlled by the SSC. The SSC transmits the data when detecting the falling edge on the TF signal after the SSC transmission is enabled. In some overflow cases, a left/right channel inversion may occur and may require SSC reinitializing.

#### Work Around

Use the SSC in Host mode so that TF is controlled by the SSC. If the SSC must be used in TF Client mode, start the SSC by writing TXEN and RXEN synchronously with the TXSYN flag rising.

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 15.2. TD output delay

The TD output is delayed by two or three extra system clock cycles when SSC is configured with the following conditions:

- RCMR.START = Start on falling edge/Start on rising edge/Start on any edge
- RFMR.FSOS = None (input)
- TCMR.START = Receive Start

#### Work Around

None

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 16. Sony/Philips Digital Interface Receiver (SPDIFRX)

### 16.1. SPDIFRX left/right inversion

The SPDIFRX can always provide the left channel sample in the first DMA buffer location when SPDIFRX\_MR.SBMODE=1. If the Soft Reset command is applied, the right channel sample may be located in the first DMA buffer location.

#### Work Around

Perform actions in the following order:

1. Disable the DMA channel.
2. Apply the Soft Reset command.
3. Perform a dummy read in the Mode register (SPDIFRX\_MR).
4. Read the SPDIFRX\_ISR.RXRDY flag. If set to 1, read the SPDIF Receiver Holding register (SPDIFRX\_RHR).

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 17. Advanced Encryption Standard (AES)

### 17.1. SPLIP mode does not work with some header sizes

The Secure Protocol Layers Improved Performances (SPLIP) mode does not work when the ESP header is not an integer multiple of 4 words.

#### Work Around

When the ESP header is not an integer multiple of 4 words, disable SPLIP mode to stop AES from automatically uploading the encrypted payload into SHA, and use the central DMA to feed SHA with the encrypted payload.

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 18. Security Module (SECUMOD)

### 18.1. Dynamic detection intrusion (PIOBU) alarm issue

The error counter fails to reinitialize after a dynamic detection intrusion (PIOBU) alarm.

#### Work Around

None

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 18.2. Tamper timestamping polarity error

The tamper detection signal polarity is inverted, with the following consequences:

- Key erasing in TZAEB, AES and TDES if the respective Clear On Tamper features are enabled, with TZAESB\_MR.TAMPCLR = 1, AES\_MR.TAMPCLR = 1 and TDES\_MR.TAMPCLR = 1.
- Scrambling key erasing in QSPI0 or QSPI1 if Clear On Tamper is enabled with QSPI0\_MR.TAMPCLR = 1 or QSPI1\_MR.TAMPCLR = 1.
- SHA locking if Tamper Lock is enabled with SHA\_MR.TMPLCK = 1. SHA is locked until SHA\_CR.UNLOCK is written to 1.

#### Work Around

Do not enable the following bits:

- TZAESB\_MR.TAMPCLR
- AES\_MR.TAMPCLR
- TDES\_MR.TAMPCLR
- QSPI0\_MR.TAMPCLR
- QSPI1\_MR.TAMPCLR
- SHA\_MR.TMPLCK

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 18.3. SECUMOD registers BMPR and WKPR reading issue

The bits 18, 19, 20 and 21 (DET0, DET1, DET2, DET3) are functional in Write mode. When they are written, the corresponding PIOBU bits are enabled as wake-up sources but, when read, the registers show those bits shifted two steps to the right, to positions 16, 17, 18 and 19, respectively.

#### Work Around

None

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 19. Gigabit Ethernet MAC (GMAC)

### 19.1. GMAC0 not functional with multiple queues in 10/100 Half Duplex mode

When operating in 10/100 Half Duplex mode, GMAC0 does not work with multiple queues.

#### Work Around

Configure the controller for transmission over a single queue.

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 19.2. Incorrect reading of Specific Address filter registers on GMAC0 and GMAC1

On GMAC0 and GMAC1, Specific Address filter register reading may be corrupted. This may impact in particular the device MAC address.

Affected registers are GMAC0\_SABx, GMAC0\_SATx, GMAC1\_SABx, GMAC1\_SATx.

#### Work Around

Use a shadow copy of the registers and fake the read operation using the shadow copy.

---

 **WARNING** With the debugger, a value read in a register may be corrupted.

---

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | -  | -      | -      | -      |  |  |  |

### 19.3. Incorrect reading of Type 1 Screener registers on GMAC0 and GMAC1

On GMAC0 and GMAC1, Type 1 Screener register reading may be corrupted. This may impact QoS-based applications using DS/TC fields or TCP/UDP ports.

Affected registers are GMAC0\_ST1RPQx and GMAC1\_ST1RPQx.

#### Work Around

Use a shadow copy of the registers and fake the read operation using the shadow copy.

---

 **WARNING** With the debugger, a value read in a register may be corrupted.

---

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | -  | -      | -      | -      |  |  |  |

### 19.4. Incorrect reading of Type 2 Screener registers on GMAC0 and GMAC1

On GMAC0 and GMAC1, Type 2 Screener register reading may be corrupted. This may impact QoS applications based on source/destination IP, source/destination TCP/UDP, Ethertype or VLAN.

Affected registers are GMAC0\_ST2RPQx, GMAC0\_ST2ERx, GMAC0\_ST2CWxRy, GMAC1\_ST2RPQx, GMAC1\_ST2ERx, GMAC1\_ST2CWxRy.

**Work Around**

Use a shadow copy of the registers and fake the read operation using the shadow copy.



**WARNING** With the debugger, a value read in a register may be corrupted.

**Affected Device Revisions**

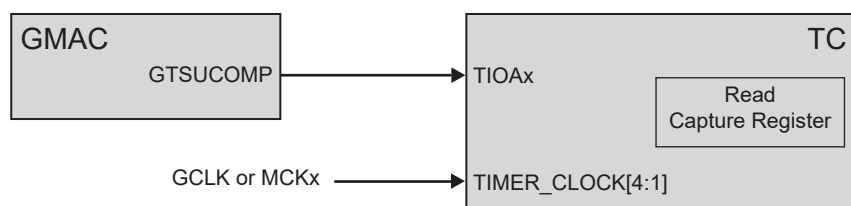
| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | –  | –      | –      | –      |  |  |  |

**19.5. GTSUCOMP ineffective connection to TC1**

The trigger/capture input B of TC1.TIOB1 is driven internally by the GTSUCOMP signal of the Ethernet MAC (GMAC), but the feature does not work.

**Work Around**

Routing the GTSUCOMP signal to one timer counter input can be done at PCB level. GTSUCOMP is multiplexed on the PB2 line, therefore PB2 can be externally looped back to one of the timer counter inputs, as shown in the following figure.

**Affected Device Revisions**

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 20. Flexible Serial Communication Controller (FLEXCOM)

### 20.1. Write Protection ineffective on FLEXCOM8 to FLEXCOM11

On FLEXCOM8, FLEXCOM9, FLEXCOM10 and FLEXCOM11, TWIx\_WPMR and TWIx\_WPSR reading may return corrupted values.

#### Work Around

Do not use the Write Protection feature on FLEXCOM8, FLEXCOM9, FLEXCOM10 and FLEXCOM11.

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 21. Secure Digital MultiMedia Card Controller (SDMMC)

### 21.1. SDMMC failure when changing speed mode or performing ALL soft reset on the fly

Changing speed mode or performing an ALL soft reset while SDCK is active may lead to block the SDMMC.

#### Work Around

Stop SDCLK before changing speed mode or performing an ALL soft reset, then re-enable SDCLK by writing SDMMC\_CCR.SDCLKEN to 0 before writing SDMMC\_MC1R/SDMMC\_HC1R/SDMMC\_HC2R.

Example:

```
//FIX : stop SDCLK
pSDMMC->SDMMC_CCR = pSDMMC->SDMMC_CCR & ~SDMMC_CCR_SDCLKEN;
switch (speed_mode) {
case DS : pSDMMC->SDMMC_HC1R = pSDMMC->SDMMC_HC1R & ~SDMMC_HC1R_HSEN;
break;
case HS : pSDMMC->SDMMC_HC1R = pSDMMC->SDMMC_HC1R | SDMMC_HC1R_HSEN;
break;
case SDR12 : pSDMMC->SDMMC_HC2R = (pSDMMC->SDMMC_HC2R & ~SDMMC_HC2R_UHSMS_Msk) |
SDMMC_HC2R_UHSMS_SDR12;
break;
case SDR25 : pSDMMC->SDMMC_HC2R = (pSDMMC->SDMMC_HC2R & ~SDMMC_HC2R_UHSMS_Msk) |
SDMMC_HC2R_UHSMS_SDR25;
break;
case SDR50 : pSDMMC->SDMMC_HC2R = (pSDMMC->SDMMC_HC2R & ~SDMMC_HC2R_UHSMS_Msk) |
SDMMC_HC2R_UHSMS_SDR50;
break;
case SDR104 : pSDMMC->SDMMC_HC2R = (pSDMMC->SDMMC_HC2R & ~SDMMC_HC2R_UHSMS_Msk) |
SDMMC_HC2R_UHSMS_SDR104;
break;
case DDR50 : pSDMMC->SDMMC_HC2R = (pSDMMC->SDMMC_HC2R & ~SDMMC_HC2R_UHSMS_Msk) |
SDMMC_HC2R_UHSMS_DDR50;
break;
}
//FIX : re-start SDCLK
pSDMMC->SDMMC_CCR = pSDMMC->SDMMC_CCR | SDMMC_CCR_SDCLKEN;
```

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 21.2. SDR104, HS200, HS400 modes are not functional

Using mode SDR104, HS200 or HS400 may lead to tuning issues, data read errors or clock switching failures.

#### Work Around

None

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 21.3. GCLK cannot be stopped

If a speed mode other than Default Speed or SDR12 is used, GCLK cannot be stopped, leading to unpredictable behavior.

### Work Around

Perform an ALL soft reset before any operation to ensure the internal clock DLL can be stopped properly.

### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 21.4. SDHC blocked after switch from high-speed mode

If the current speed mode is Default Speed or SDR12, the SDCLK frequency must be higher than 25 MHz before switching to another mode, otherwise SDHC is blocked.

### Work Around

Set the SDCLK frequency to a value higher than 25 MHz before switching to another mode.

### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 22. Controller Area Network (MCAN)

### 22.1. Edge filtering causes mis-synchronization when falling edge at Rx input pin coincides with end of integration phase



**Attention:** This erratum is not relevant for CAN 2.0.

When edge filtering is enabled (MCAN\_CCCR.EFBI = '1') and when the end of the integration phase coincides with a falling edge at the Rx input pin, it may happen that the MCAN synchronizes itself wrongly and does not correctly receive the first bit of the frame. In this case the CRC will detect that the first bit was received incorrectly; it will rate the received FD frame as faulty and an error frame will be sent.

The issue only occurs when there is a falling edge at the Rx input pin (CANRX) within the last time quantum (tq) before the end of the integration phase. The last time quantum of the integration phase is at the sample point of the 11th recessive bit of the integration phase. When the edge filtering is enabled, the bit timing logic of the MCAN sees the Rx input signal delayed by the edge filtering. When the integration phase ends, the edge filtering is automatically disabled. This affects the reset of the FD CRC registers at the beginning of the frame. The Classical CRC registers are not affected, so this issue does not affect the reception of Classical frames.

In CAN communication, the MCAN may enter integrating state (either by resetting MCAN\_CCCR.INIT or by protocol exception event) while a frame is active on the bus. In this case the 11 recessive bits are counted between the Acknowledge bit and the following start of frame. All nodes have synchronized at the beginning of the dominant Acknowledge bit. This means that the edge of the following Start-of-Frame bit cannot fall on the sample point, so the issue does not occur. The issue occurs only when the MCAN is, by local errors, mis-synchronized with regard to the other nodes, or not synchronized at all.

Glitch filtering as specified in ISO 11898-1:2015 is fully functional.

Edge filtering was introduced for applications where the data bit time is at least two tq (of the nominal bit time) long. In that case, edge filtering requires at least two consecutive dominant time quanta before the counter counting the 11 recessive bits for idle detection is restarted. This means edge filtering covers the theoretical case of occasional 1-tq-long dominant spikes on the CAN bus that would delay idle detection. Repeated dominant spikes on the CAN bus would disturb all CAN communication, so the filtering to speed up idle detection would not help network performance.

When this rare event occurs, the MCAN sends an error frame and the sender of the affected frame retransmits the frame. When the retransmitted frame is received, the MCAN has left the integration phase and the frame will be received correctly. Edge filtering is only applied during integration phase; it is never used during normal operation. As the integration phase is very short with respect to "active communication time", the impact on total error frame rate is negligible. The issue has no impact on data integrity.

The MCAN enters integration phase under the following conditions:

- when MCAN\_CCCR.INIT is set to '0' after start-up
- after a protocol exception event (only when MCAN\_CCCR.PXHD = '0')

#### Work Around

Disable edge filtering or wait on retransmission in case this rare event happens.

**Affected Device Revisions**

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

**22.2. Configuration of MCAN\_NBTP.NTSEG2 = '0' not allowed**

**Attention:** This erratum is applicable for CAN 2.0.

When MCAN\_NBTP.NTSEG2 is configured to zero (Phase\_Seg2(N) = 1), and when there is a pending transmission request, a dominant third bit of Intermission may cause the MCAN to wrongly transmit the first identifier bit dominant instead of recessive, even if this bit was configured as '1' in the MCAN's Tx Buffer Element.

A phase buffer segment 2 of length '1' (Phase\_Seg2(N) = 1) is not sufficient to switch to the first identifier bit after the sample point in Intermission where the dominant bit was detected.

The CAN protocol according to ISO 11898-1 defines that a dominant third bit of Intermission causes a pending transmission to be started immediately. The received dominant bit is handled as if the MCAN has transmitted a Start-of-Frame (SoF) bit.

The ISO 11898-1 specifies the minimum configuration range for Phase\_Seg2(N) to be 2..8 tq. Therefore excluding a Phase\_Seg2(N) of '1' will not affect MCAN conformance.

**Work Around**

Use the range 1..127 for MCAN\_NBTP.NTSEG2 instead of 0..127.

**Affected Device Revisions**

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

**22.3. Retransmission in DAR mode due to lost arbitration at the first two identifier bits**

**Attention:** This erratum is applicable for CAN 2.0.

When the MCAN is configured in DAR mode (MCAN\_CCCR.DAR = '1') the Automatic Retransmission for transmitted messages that have been disturbed by an error or have lost arbitration is disabled. When the transmission attempt is not successful, the Tx Buffer's transmission request bit (MCAN\_TXBRP.TRPxx) shall be cleared and its Cancellation Finished bit (MCAN\_TXBCF.CFxx) shall be set.

When the transmitted message loses arbitration at one of the first two identifier bits, it may happen that instead of the bits of the actually transmitted Tx Buffer, the MCAN\_TXBRP.TRPxx and MCAN\_TXBCF.CFxx bits of the previously started Tx Buffer (or Tx Buffer 0 if there is no previous transmission attempt) are written (MCAN\_TXBRP.TRPxx = '0', MCAN\_TXBCF.CFxx = '1').

If in this case the MCAN\_TXBRP.TRPxx bit of the Tx Buffer that lost arbitration at the first two identifier bits has not been cleared, retransmission is attempted.

When the MCAN loses arbitration again at the immediately following retransmission, then actually and previously transmitted Tx Buffers are the same and this Tx Buffer's MCAN\_TXBRP.TRPxx bit is cleared and its MCAN\_TXBCF.CFxx bit is set.

**Work Around**

None

**Affected Device Revisions**

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

**22.4. Tx FIFO message sequence inversion****Attention:** This erratum is applicable for CAN 2.0.

Assume the case that there are two Tx FIFO messages in the output pipeline of the Tx Message Handler. Transmission of Tx FIFO message 1 is started:

- Position 1: Tx FIFO message 1 (transmission ongoing)
- Position 2: Tx FIFO message 2
- Position 3: --

Now a non-Tx FIFO message with a higher CAN priority is requested. Due to its priority it will be inserted into the output pipeline. The TxMH performs so called "message scans" to keep the output pipeline up to date with the highest priority messages from the Message RAM. After the following two message scans, the output pipeline has the following content:

- Position 1: Tx FIFO message 1 (transmission ongoing)
- Position 2: non-Tx FIFO message with higher CAN priority
- Position 3: Tx FIFO message 2

If the transmission of Tx FIFO message 1 is not successful (lost arbitration or CAN bus error) it is pushed from the output pipeline by the non-Tx FIFO message with higher CAN priority. The following scan re-inserts Tx FIFO message 1 into the output pipeline at position 3:

- Position 1: non-Tx FIFO message with higher CAN priority (transmission ongoing)
- Position 2: Tx FIFO message 2
- Position 3: Tx FIFO message 1

Now Tx FIFO message 2 is in the output pipeline in front of Tx FIFO message 1 and they are transmitted in that order, resulting in a message sequence inversion.

**Work Around****1. First Work Around**

Use two dedicated Tx Buffers, e.g. use Tx Buffers 4 and 5 instead of the Tx FIFO. The pseudo-code below replaces the function that fills the Tx FIFO.

Write message to Tx Buffer 4.

Transmit loop:

- Request Tx Buffer 4 - write MCAN\_TXBAR.A4
- Write message to Tx Buffer 5
- Wait until transmission of Tx Buffer 4 completed - MCAN\_IR.TC, read MCAN\_TXBTO.TO4
- Request Tx Buffer 5 - write MCAN\_TXBAR.A5
- Write message to Tx Buffer 4
- Wait until transmission of Tx Buffer 5 is completed - MCAN\_IR.TC, read MCAN\_TXBTO.TO5

## 2. Second Work Around

Make sure that only one Tx FIFO element is pending for transmission at any time. The Tx FIFO elements may be filled at any time with messages to be transmitted, but their transmission requests are handled separately. Each time a Tx FIFO transmission has completed and the Tx FIFO gets empty (MCAN\_IR.TFE = '1'), the next Tx FIFO element is requested.

## 3. Third Work Around

Use only a Tx FIFO. Send the message with the higher priority also from Tx FIFO.

One drawback is that the higher priority message has to wait until the preceding messages in the Tx FIFO have been sent.

### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 22.5. Unexpected High Priority Message (HPM) interrupt



**Attention:** This erratum is applicable for CAN 2.0.

This issue occurs in two configurations:

### Configuration A:

- At least one Standard Message ID Filter Element is configured with Priority flag set (S0.SFEC = "100"/"101"/"110").
- No Extended Message ID Filter Element is configured.
- Non-matching extended frames are accepted (MCAN\_GFC.ANFE = "00"/"01").

The HPM Interrupt flag MCAN\_IR.HPM is set erroneously on reception of a non-high-priority extended message under the following conditions:

1. A standard HPM frame is received, and accepted by a filter with Priority flag set. Then, Interrupt flag MCAN\_IR.HPM is set as expected.
2. Next, an extended frame is received and accepted due to the MCAN\_GFC.ANFE configuration. Then, Interrupt flag MCAN\_IR.HPM is set erroneously.

### Configuration B:

- At least one Extended Message ID Filter Element is configured with Priority flag set (F0.EFEC = "100"/"101"/"110").
- No Standard Message ID Filter Element is configured.
- Non-matching standard frames are accepted (MCAN\_GFC.ANFS = "00"/"01").

The HPM Interrupt flag MCAN\_IR.HPM is set erroneously on reception of a non-high-priority standard message under the following conditions:

1. An extended HPM frame is received, and accepted by a filter with Priority flag set. Then, Interrupt flag MCAN\_IR.HPM is set as expected.
2. Next, a standard frame is received and accepted due to the MCAN\_GFC.ANFS configuration. Then, Interrupt flag MCAN\_IR.HPM is set erroneously.

### Work Around

#### Configuration A:

Set up an Extended Message ID Filter Element with the following configuration:

- F0.EFEC = "001"/"010" - select Rx FIFO for storage of extended frames
- F0.EFID1 = any value - value not relevant as all ID bits are masked out by F1.EFID2
- F1.EFT = "10" - classic filter, F0.EFID1 = filter, F1.EFID2 = mask
- F1.EFID2 = zero - all bits of the received extended ID are masked out

Now, all extended frames are stored in Rx FIFO 0 respectively Rx FIFO 1 depending on the configuration of F0.EFEC.

#### Configuration B:

Set up a Standard Message ID Filter Element with the following configuration:

- S0.SFEC = "001"/"010" - select Rx FIFO for storage of standard frames
- S0.SFID1 = any value - value not relevant as all ID bits are masked out by S0.SFID2
- S0.SFT = "10" - classic filter, S0.SFID1 = filter, S0.SFID2 = mask
- S0.SFID2 = zero - all bits of the received standard ID are masked out

Now, all standard frames are stored in Rx FIFO 0 respectively Rx FIFO 1 depending on the configuration of S0.SFEC.

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 22.6. Issue message transmitted with wrong arbitration and control fields



**Attention:** This erratum is applicable for CAN 2.0.

When the following conditions are met, a message with wrong ID, format, and DLC is transmitted:

- M\_CAN is in state "Receiver" (PSR.ACT = "10") and there is no pending transmission.
- A new transmission is requested before the third Intermission bit is reached.
- The CAN bus is sampled dominant at the third Intermission bit which is treated as SoF (see ISO11898-1:2015 Section 10.4.2.2).

Then, it can happen that:

- The Shift register is not loaded with the ID, format and DLC of the requested message.
- The MCAN starts arbitration with wrong ID, format, and DLC on the next bit.
- If the ID wins arbitration, a CAN message with valid CRC is transmitted.
- If this message is acknowledged, the ID stored in the Tx Event FIFO is the ID of the requested Tx message, and not the ID of the message transmitted on the CAN bus, and no error is detected by the transmitting MCAN.

#### Work Around

Request a new transmission only if another transmission is already pending or when the MCAN is not in "Receiver" state (when PSR.ACT ≠ "10").

To avoid activating the transmission request in the critical time window between the sample points of the second and third Intermission bits, the application software can evaluate the Rx Interrupt flags IR.DRX, IR.RF0N and IR.RF1N, which are set at the last EoF bit when a received and accepted message becomes valid.

The last EoF bit is followed by three Intermission bits. Therefore, the critical time window has safely terminated three bit times after the Rx interrupt. Now a transmission can be requested by writing to TXBAR.

After the interrupt, the application has to take care that the transmission request for the CAN Protocol Controller is activated before the critical window of the following reception is reached.

A checksum covering the arbitration and control fields can be added to the data field of the message to be transmitted, to detect frames transmitted with wrong arbitration and control fields.

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 22.7. Debug message handling state machine not reset to Idle when CCCR.INIT is set

When the host sets the MCAN\_CCCR.INIT bit through the MCAN\_CCCRn register, or when the CAN enters Bus Off state, the debug message handling state machine stays in its current state instead of resetting to Idle state. Setting MCAN\_CCCR.CCE does not change MCAN\_RXF1S.DMS.

#### Work Around

If the debug message handling state machine stopped while MCAN\_RXF1S.DMS="01" or MCAN\_RXF1S.DMS="10", it can be reset to Idle state by hardware reset or by reception of debug messages after MCAN\_CCCR.INIT is reset to zero.

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 22.8. Message order inversion when transmitting from dedicated Tx buffers configured with same message ID

When several Tx buffers are configured with the same message ID, transmission of these Tx buffers is requested sequentially with a delay between the individual Tx requests. They are transmitted in ascending order of their numbers, so the Tx buffer with the lowest number and pending Tx request is transmitted first.

However, depending on the delay between the individual Tx requests, it can happen that the lowest Tx buffer number is not transmitted first and that the message order is inverted.

#### Work Around

First write the group of Tx messages having the same message ID to the message RAM, and then request transmission of all these messages concurrently by a single write access to MCAN\_TXBAR.

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

### 22.9. Frame transmitted despite confirmed transmit cancellation for CAN-FD messages with more than 8 data bytes



**Attention:** This erratum is not relevant for CAN 2.0.

In case the transmission of Tx Buffer nn was not successful and is restarted immediately afterwards by automatic retransmission, and the software requests a Tx cancellation for this Tx Buffer by

setting the cancellation request bit MCAN\_TXBCR.CRnn during transmission of the first 4 identifier bits, a successful cancellation is incorrectly signalled by setting MCAN\_TXBCF.CFnn = '1' and by clearing MCAN\_TXBRP.TRPnn. In addition, the respective Transmission Occurred bit remains zero (MCAN\_TXBTO.TOnn = '0'), incorrectly indicating that the frame was not transmitted on the bus.

Other than signalled by MCAN\_TXBCF.CFnn and MCAN\_TXBTO.TOnn, the transmission continues until the complete frame has been sent on the CAN bus. If the transmission is successful, MCAN\_TXBTO.TOnn will be set.

If in this case new data is written to Tx Buffer nn while the transmission is still ongoing, a frame with inconsistent data may appear on the bus.

This problem is limited to the case of transmit cancellation of CAN-FD messages with more than 8 data bytes while automatic retransmission is enabled (MCAN\_CCCR.DAR = '0').

Transmit cancellation of Classical CAN messages and CAN-FD messages with up to 8 data bytes is not affected.

CAN 2.0 operation is not impacted.

### Work Around

Do not use transmit cancellation for CAN-FD messages with more than 8 data bytes.

Alternatively, wait for the duration of the expected transmission time of the cancelled Tx Buffer before writing new data to that Tx Buffer.

### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 22.10. MCAN\_TSU\_TSCFG reset after read

When a write is issued to configure the TSU Timestamp Configuration register (MCAN\_TSU\_TSCFG), any attempt to read it resets the register content.

### Work Around

Save the content of MCAN\_TSU\_TSCFG in memory to access the value without reading the register.

### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 22.11. MCAN\_TSU\_TSS1 not reset after a MCAN\_TSU\_TSx read

TSL[15:0] and TSN[15:0] in MCAN TSU Timestamp Status 1 (MCAN\_TSU\_TSS1) are not reset after reading a MCAN TSU Timestamp Status (MCAN\_TSU\_TSx) register.

### Work Around

Proceed as follows:

1. Read MCAN\_TSU\_TSx.
2. For each bit set, read the corresponding MCAN\_TSU\_TSx and save the values.
3. Write the same MCAN\_TSU\_TSx register with value '0' to reset the corresponding bit in MCAN\_TSU\_TSS1.

### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 22.12. MCAN\_TSU\_ATB read resets the timebase value

Each access to the Actual Timebase register (MCAN\_TSU\_ATB) resets the Timebase Prescaler MCAN\_TSU\_TSCFG.TBPRE.

### Work Around

None

### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 23. Timer Counter (TC)

### 23.1. TC0 Channel 2 registers incorrect reading

Reading the following registers (TC0 Channel 2 registers) may return corrupted values: TC0\_CCR2, TC0\_CMR2, TC0\_SMMR2, TC0\_RAB2, TC0\_CV2, TC0\_RA2, TC0\_RB2, TC0\_RC2, TC0\_SR2, TC0\_IER2.

#### Work Around

Do not use TC0 Channel 2.

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | -  | -      | -      | -      |  |  |  |

## 24. USB Device High Speed Port (UDPHS)

### 24.1. EHCI spurious stop when Suspend mode occurs on port A

If port A enters Suspend mode (via the USB device controller or the USB host controller), the EHCI clock stops, which can block the EHCI if it uses the other ports (B/C). This occurs after about 2  $\mu$ s ( $\approx$  120 clock cycles).

#### Work Around

Set the SFR\_EHCI0HCI.PHYCLK bit so that the EHCI clock remains activated in Suspend mode, with no power saving.

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 25. Low Power Modes

### 25.1. ULP2 mode does not work

Operation of the device at ULP2 wake-up is not predictable. Do not use this low-power mode.

#### Work Around

None

#### Affected Device Revisions

| A0 | A1 | A1-D1G | A1-D2G | A1-D4G |  |  |  |
|----|----|--------|--------|--------|--|--|--|
| X  | X  | X      | X      | X      |  |  |  |

## 26. Data Sheet Clarifications

### 26.1. SAMA7G5 Series Data Sheet

The following clarifications are applicable to all SAMA7G5 Series data sheet revisions, up to and including revision D (DS60001765D).

#### 26.1.1. Octal Flash boot supported in specific mode only

In addition to the information provided in the "QSPI NOR Flash Boot" section of the device data sheet, consider the following:

The ROM code supports booting out of octal SPI memories only in SPI mode (1-bit). Quad and Octal modes are not supported.

#### 26.1.2. FLEXCOM USART Timings update

The following table supersedes the "FLEXCOM USART Timings" table in the Electrical Characteristics section of the data sheet. This new table allows for greater USART speeds in Asynchronous mode.

**Table 26-1.** FLEXCOM USART Timings

| Symbol    | Parameter                                                              | Conditions          | Min | Max             | Unit |
|-----------|------------------------------------------------------------------------|---------------------|-----|-----------------|------|
| $f_{SCK}$ | SCK frequency in Synchronous mode<br>(FLEX_US_MR.SYNC = 1)             | Host or client      | –   | 25              | MHz  |
| BR        | Baud rate in Asynchronous mode <sup>(1)</sup><br>(FLEX_US_MR.SYNC = 0) | FLEX_US_MR.OVER = 1 | –   | $f_{MCK1} / 8$  | Mbps |
|           |                                                                        | FLEX_US_MR.OVER = 0 | –   | $f_{MCK1} / 16$ |      |

**Note:**

1. With  $f_{MCK1} = 200$  MHz, the maximum baud rate is:

BR =  $200 / 8 = 25$  Mbps (OVER = 1), or

BR =  $200 / 16 = 12.5$  Mbps (OVER = 0)

#### 26.1.3. Selected I2SMCC flags are not cleared on read

In the I2SMCC Interrupt Status register B (I2SMCC\_ISRB), the TXFFRDY, TXFFEMP, RXFFRDY and RXFFFUL FIFO status flags are incorrectly described as being cleared on read.

The correct clear conditions for these flags are as follows:

- TXFFRDY is cleared when the transmit FIFO becomes full.
- TXFFEMP is cleared when at least one data word is written to I2SMCC\_THR.
- RXFFRDY is cleared when there is no more data available to read from I2SMCC\_RHR.
- RXFFFUL is cleared when one data word is read from I2SMCC\_RHR.

### 26.2. SAMA7G5 Series SiP Data Sheet

Clarifications listed in the [SAMA7G5 Series Data Sheet](#) section also apply to the SAMA7G5 Series SiP devices.

There are no other clarifications to report.

## 27. Revision History

### 27.1. DS80001016J - 09/2026

- [Table 1](#): updated part numbers from SAMA7G54DxG-I/XHB to SAMA7G54DxG-V/XHB
- In [CSI-2 Demultiplexer Controller \(CSI2DC\)](#), added [Decompression modes 12-7-12, 12-8-12, 10-7-10 and 10-6-10 not compliant](#)
- In [Data Sheet Clarifications](#), added [Selected I2SMCC flags are not cleared on read](#)
- Removed the following errata:
  - NAND Flash boot not supported
  - OTPC limited number of packets

### 27.2. DS80001016I - 03/2026

Extended scope to the SAMA7G54DxG(T)-I/XHB and SAMA7G54DxG(T)-E/XHB(VAO) devices

### 27.3. DS80001016H - 01/2026

- Updated “NAND Flash boot not supported” erratum
- In [Data Sheet Clarifications](#), added:
  - [Octal Flash boot supported in specific mode only](#)
  - [FLEXCOM USART Timings update](#)

### 27.4. DS80001016G - 09/2025

Extended scope to the SAMA7G54(T)-V/7EW device

### 27.5. DS80001016F - 06/2025

Extended scope to the SAMA7G54D4G(T)-I/4UB device  
 Removed SAMA7G54(T)-V/7EW from scope  
[Table 1](#): corrected ordering codes from SAMA7G54DxG(T)-I/4TB to SAMA7G54DxG(T)-I/4UB

### 27.6. DS80001016E - 03/2025

- Throughout:
  - Extended scope to the SAMA7G54(T)-V/7EW device
  - Changed terminology from “silicon revision” to “device revision”
- Added [Boot failure on eMMC memories](#)

### 27.7. DS80001016D - 11/2024

Added:

- [MCAN\\_TSU\\_TSCFG reset after read](#)
- [MCAN\\_TSU\\_TSS1 not reset after a MCAN\\_TSU\\_TSx read](#)
- [MCAN\\_TSU\\_ATB read resets the timebase value](#)
- [ULP2 mode does not work](#)

Updated [MCKRDY flag error](#), [GTSUCOMP ineffective connection to TC1](#)

## 27.8. DS80001016C - 12/2023

Added references to SAMA7G5 Series SiP devices.

Added:

- [CHIPID\\_EXID may report a wrong value](#)
- [Delay to first establish PCK](#)
- [PCK and GCLK Ready status issue](#)
- [Processor \(CPU\\_CLK0\) and main system bus clock \(MCK0\) source selection](#)
- [SPLIP mode does not work with some header sizes](#)
- [SECUMOD registers BMPR and WKPR reading issue](#)

Updated [MCKRDY flag error](#)

Removed "Temperature sensor wrong parameter value" data sheet clarification

## 27.9. DS80001016B - 08/2022

Added:

- [Temperature sensor still enabled when stopped without conversion](#)
- [Temperature sensor spurious activation with CH30](#)
- [Sleep mode ineffective](#)
- [Frame transmitted despite confirmed transmit cancellation for CAN-FD messages with more than 8 data bytes](#)
- "Temperature sensor wrong parameter value" in [Data Sheet Clarifications](#)

Updated [Spurious DMA descriptor writing](#)

Removed ACC erratum "WPVS bit incorrect behavior"

## 27.10. DS80001016A - 03/2022

First issue.

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## Product Page Links

[SAMA7G54](#), [SAMA7G54D1G](#), [SAMA7G54D2G](#), [SAMA7G54D4G](#)